Retrieval that Cannot See the Future: Temporal Evaluation of Fact-Checked Claim Retrieval Project Proposal for NLP Course, Winter 2026/27

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Abstract

Previously fact-checked claim retrieval (PFCR) ranks an archive of fact-checks for a social media post. Published benchmarks search the full archive, including fact-checks written after the post, while a fact-checker at post time can only search what already exists. We propose a *causal* evaluation protocol: for each post q with time t_q, the pool is $P(q) = \{f : t_f < t_q\}$, and we report Success@10 (fraction of queries with a gold fact-check in the top 10) and mean reciprocal rank (MRR) under random, size-matched random, and strict-causal pools on MultiClaim, with Polish as focus and English as control. Deliverables: the protocol and rebuildable split as id lists and code; a results table of at least four systems under both protocols; and a gap decomposition with 50 hand-coded newly failing queries.

1 Scientific goal

Problem. PFCR matches a post q to earlier fact-checks f that already debunk the same claim (Shaar et al., 2020; Pikuliak et al., 2023). Standard evaluation lets q retrieve fact-checks published after t_q, which rewards matching a later article that quotes the post. We ask how much reported performance survives a time-respecting index, and why it drops.

Research questions. RQ1: How much Success@10 (S@10) and mean reciprocal rank (MRR) survive a causal index (unconditionally and on answerable queries), and does system ranking change? **RQ2:** How much of the random-vs-causal gap is answerability loss, pool-size effects, training leakage, or missing labels for earlier sibling debunks? **RQ3:** Is the causal penalty larger for Polish than for English, and how much do earlier English fact-checks recover for Polish posts? **RQ4:** Do large language model (LLM) rerankers gain on facts they may have seen in pretraining (core+, after the committed core)?

Hypotheses (pre-registered). Most MultiClaim gold pairs are post-before-fact-check (H1); on answerable queries, time-local distractors are harder than a size-matched random pool (H2); Polish shows a larger causal penalty than English, especially for lexical BM25 (H3); fine-tuned retrievers lose more under chronological training than zero-shot ones (Shaar et al., 2022) (H4).

2 Significance and state of the art

We summarise fifteen prior works. Table 1 lists the published numbers we compare, with the dataset and split named for every value. No PFCR evaluation we found restricts the fact-check index *per query* to documents published before the post.

PFCR task and shared tasks. (Shaar et al., 2020) define PFCR as ranking verified claims for a social-media or debate input; on Snopes tweets and PolitiFact debate sentences, BM25 plus rankSVM reaches MRR 0.788 and S@10 0.915 on the Snopes test (BM25 top-50 rerank). (Shaar et al., 2021) run CLEF CheckThat! 2021 Task 2 (English/Arabic tweets; English debates) with fixed pools and no date filter; best English tweet MAP@5 is 0.883 (Aschern) vs BM25 0.749 on the 70/15/15 tweet split. (Nakov et al., 2022) rerun the lab in 2022 with a small new test set; RIET Lab reaches MAP@5 0.956 on English

tweets (2022 test) with sentence-T5 plus GPT-Neo reranking. (Hardalov et al., 2022) mine 333k noisy tweet–Snopes pairs from crowd replies and train sentence-BERT; on CheckThat! 2021 English test they report MAP@5 90.3 for the full system (vs Aschern 88.2).

Multilingual PFCR corpora and systems. (Pikuliak et al., 2023) release MultiClaim (28k posts, 206k fact-checks, 31k pairs) and evaluate unsupervised and fine-tuned retrievers over the *full* pool; best monolingual S@10 is 0.83 (GTR-T5-Large on English translations). Their post-newer vs post-older subsets score 80.02 vs 80.04 S@10 under the same full index (no per-query date filter). (Peng et al., 2025) organise SemEval-2025 Task 7 on MultiClaim with out-of-time test posts (pool to May 2024); winner PINGAN AI reaches monolingual average S@10 0.9601 and Polish 0.9260 on the hidden test languages. (Devadiga et al., 2025) (SemEval system) translate posts, fine-tune Stella with hard negatives and LLM-rerank; monolingual test S@10 0.938 (Polish 0.886). (Ramponi et al., 2025) compare multilingual vs crosslingual retrieval on a curated MultiClaim v2 subset (47 languages) with an 80/10/10 language-stratified split that separates fact-checks across folds; LLM reranking of multilingual-e5-large reaches MRR@10 0.8042 multilingual. (Vykopal et al., 2025) release AMC-16K (16k human Yes/No pairs from MultiClaim; 16% relevant overall) and evaluate LLMs as relevance detectors; best Macro-F1 82.46 monolingual (Mistral Large few-shot).

Closest temporal precedents for PFCR. (Shaar et al., 2022) add transcript context and a debate-level chronological train/test split on PolitiFact; the prior SOTA reranker drops from MAP 0.602 (sentence-level random) to 0.429 (debate chrono) without restricting the claim pool by date.

Time-respecting IR and verification (not PFCR index filtering). (Ounis et al., 2011) define TREC 2011 Microblog real-time search with a compulsory run that bans future tweets and future corpus statistics; best compulsory P@30 is 0.4551. (Liška et al., 2022) evaluate QA over a growing dated news corpus by quarter; dated DPR raises recall@20 on recent generated questions from 43% to 57%. (Glockner et al., 2022) show that 69.7% of MultiFC misinformation claims have leaked fact-check evidence in search snippets (verification leakage from later fact-checks). (Schlichtkrull et al., 2023) release AVeriTeC with evidence restricted to documents before the claim and temporally ordered splits; baseline verdict macro-F1 is 0.23 (gold evidence 0.49) on the AVeriTeC test. (Mu et al., 2023) show rumour classifiers collapse under chronological splits: on Twitter 16, BERT macro-F1 falls from 91.5 (random) to 47.2 (stratified chronological).

Table metrics. Let $G(q)$ be the gold items for query q and r_q the rank of the first gold hit (undefined if none). *S@10* is the fraction of queries with a gold hit in the top 10; *MRR* is the mean of $1/r_q$ (0 if no gold ranked) (Pikuliak et al., 2023; Peng et al., 2025). *MRR@10* clips reciprocal rank at 10. *MAP@k* averages, over queries, precision at each relevant hit within the top k (plain MAP uses the full list). *P@k / R@k*: fraction of the top k that are relevant / fraction of all gold recovered in the top k. *Macro-F1*: unweighted mean of per-class F1 (classification, not ranking). *Leak rate*: share of claims whose later fact-check evidence already appears in search snippets (Glockner et al., 2022).

3 Datasets and status

Table 2 lists the source datasets (derived annotations such as failure codes are omitted here). For ranking corpora, class balance is the density of positive post–fact-check links, not a balanced classification split.

4 Concept, methodology and work plan

Pipeline. Figure 1 summarises the evaluation path. Systems (committed core): BM25; multilingual-e5-large (Wang et al., 2024); a Polish or cross-lingual encoder (mmlw / LaBSE); bge-reranker-v2-m3 on the top 100 (Chen et al., 2024). Protocols: full-pool random; strict causal $P(q)$; size-matched random of $|P(q)|$. Compute: local open-weight models only; no commercial LLM API on post text.

Metrics. Chosen before results (MultiClaim / SemEval-2025 convention (Pikuliak et al., 2023; Peng et al., 2025)). Let $G(q)$ be the gold fact-checks for query q and r_q the rank of the first gold hit ($1/r_q = 0$ if none). *S@10* is the fraction of queries with $r_q \leq 10$; *MRR* is the mean of $1/r_q$. *Answerable rate* is the share of queries with $G(q) \cap P(q) \neq \emptyset$ under the causal pool. We report

Table 1: Published numbers for the fifteen works above. Every value names the dataset and the split or evaluation condition used in the source paper. Metrics as above.

Work	Metric	Value	Dataset / split
(Shaar et al., 2020)	MRR	0.788	Snopes tweets / official test; BM25 top-50 + rankSVM
(Shaar et al., 2020)	S@10	0.915	same
(Shaar et al., 2021)	MAP@5	0.883	CheckThat! 2021 EN 2A / 70-15-15 test (Aschern)
(Shaar et al., 2021)	MAP@5	0.749	same / BM25 baseline
(Nakov et al., 2022)	MAP@5	0.956	CheckThat! 2022 EN 2A / 2022 test (RIET Lab)
(Hardalov et al., 2022)	MAP@5	90.3	CheckThat! 2021 EN test / CrowdChecked-trained system
(Pikuliak et al., 2023)	S@10	0.83	MultiClaim / full pool, mono, GTR-T5-L (EN text)
(Pikuliak et al., 2023)	S@10	80.02 / 80.04	MultiClaim / post-newer vs older, full index, GTR-T5
(Peng et al., 2025)	S@10	0.9601	SemEval-2025 / mono test avg (PINGAN AI)
(Peng et al., 2025)	S@10	0.926	SemEval-2025 / Polish test (PINGAN AI)
(Devadiga et al., 2025)	S@10	0.938	SemEval-2025 / mono test (TIFIN)
(Devadiga et al., 2025)	S@10	0.886	SemEval-2025 / Polish test (TIFIN)
(Ramponi et al., 2025)	MRR@10	0.804	MultiClaim v2 curated / 80-10-10 + LLM rerank
(Vykopal et al., 2025)	Macro-F1	82.46	AMC-16K / mono human labels, Mistral few-shot
(Shaar et al., 2022)	MAP	0.602	PolitiFact debates / sentence-level random
(Shaar et al., 2022)	MAP	0.429	PolitiFact debates / debate-level chronological
(Ounis et al., 2011)	P@30	0.455	TREC 2011 Microblog / compulsory no-future run
(Liška et al., 2022)	R@20	57% / 43%	StreamingQA / 2020 recent gen. Qs; dated vs undated DPR
(Glockner et al., 2022)	leak rate	69.7%	MultiFC misinfo claims / auto leak detector
(Schlichtkrull et al., 2023)	macro-F1	0.23 / 0.49	AVeriTeC test / baseline vs gold evidence
(Mu et al., 2023)	macro-F1	91.5 / 47.2	Twitter 16 / random vs stratified chronological, BERT

both unconditionally (unanswerable queries count as misses) and on answerable queries only; significance via paired cluster bootstrap; Polish–English contrast via difference-in-differences.

Concrete success deliverables. (1) Rebuildable causal split (id lists + code; no post text). (2) Shared results table: $\geq$4 systems $\times$ random vs causal $\times$ Polish and English, with bootstrap intervals; central number = extra Polish temporal cost. (3) Gap decomposition plus coded analysis of 50 newly failing queries.

Week-by-week plan. Internal work finishes before Christmas; course checkpoints: mid-term video 18 Nov, Phase 2 2 Dec, formal final 13 Jan 2027.

- **W1 (to 18 Nov):** download MultiClaim v1/v2 and SemEval-2025 Task 7, dated corpus, protocol module, answerable-rate gate. Owner: Paweł Pozorski.
- **W2 (to 2 Dec):** core systems (BM25, dense, reranker) for Polish and English; Phase 2 PoC. Owner: Mateusz Jarosz (with Paweł Florek on GPU runs).
- **W3 (to 16 Dec):** bootstrap tables, gap decomposition, 50 failure annotations. Owner: Paweł Florek (with Jan Cwalina on annotation).
- **W4 (to 23 Dec):** release check, write-up; core+ only if time remains. Project deliverables done before Christmas. Owner: Jan Cwalina (with Paweł Pozorski on writing).

Main risks: unknown Polish answerable rate (if $<$150, Polish–English contrast becomes descriptive); timestamp semantics undocumented (URL date audit + margin sensitivity); AMC-16K still to be requested.

5 Data statement

Primary data: MultiClaim v1/v2 and SemEval-2025 Task 7 from restricted Zenodo records (research use only; no re-identification or re-sharing beyond named requesters) (Moro et al., 2025; Pikuliak et al., 2023; Peng et al., 2025); MultiClaimNet (Zenodo; CC BY 4.0) (Panchendrarajan et al., 2025). Access is granted for MultiClaim v1/v2, SemEval-2025 Task 7 and MultiClaimNet. Posts are real social-media text, released in anonymised form by the dataset authors; we treat them as personal data. We store files only in a directory readable by named team members, never send post text to an external LLM API, never put raw text in git or the paper (examples paraphrased; errors aggregated), and delete the data at project end. AMC-16K is secondary (LLM-judge validation only). Its public listing is on the EU

Table 2: Source datasets. FC = fact-checks: short verified/debunked claims from fact-checking sites (the documents we retrieve).

Dataset	Size	Label definition	Class balance	Licence	How obtained	Status
MultiClaim v1	206k FC, 28k posts, 31k pairs	fact-checker gold links (ClaimReview / warnings)	sparse positives (ranking)	research use only	Zenodo 7737983	READY
MultiClaim v2	435k FC, 89k posts, 105k pairs	same + relationship types	sparse positives (ranking)	research use only	Zenodo 15413169	READY
SemEval-2025 T7	train from MultiClaim; test 4.3k mono posts (PL: 500 / 564 pairs)	shared-task gold for S@10	sparse positives (ranking)	research use only	Zenodo 14989176	READY
MultiClaimNet	85.3k FC in 30.9k clusters	same-claim cluster ids	n/a (clustering)	CC BY 4.0	Zenodo 15100352	READY
AMC-16K	16k pairs / 20 langs	human Yes/No relevance	16% relevant overall; PL mono 20.25%	CC BY-NC-SA 4.0	authors (not via LDS)	RESTRICTED – to be requested

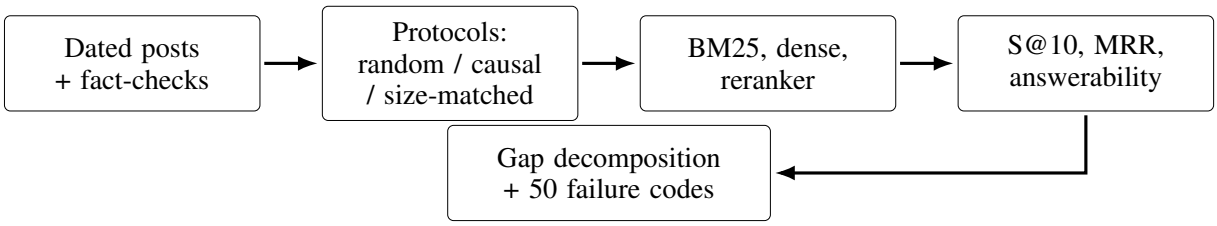


Figure 1: Evaluation pipeline.

Table 3: Contributions to preparing this report. Hours are approximate.

Member	Role focus	Hours
Paweł Pozorski	data, protocol & SOTA	7
Mateusz Jarosz	systems & SOTA	7
Paweł Florek	evaluation, figure & LaTeX	5
Jan Cwalina	drafting & polish	5

Language Data Space catalogue, which WUT cannot use; we will request a copy from the authors (KInIT) under research terms.

6 Contribution

Table 3 lists each member's tasks and approximate hours spent preparing this report.

7 LLM-use declaration

We used AI assistants while preparing this proposal and report. Their role was limited to formatting (LaTeX tables, section structure, citation layout) and wording corrections (clarity, grammar, consistency of terminology).

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